

I2C Driver Analysis

Prepared By: Anagha

Introduction

I2C is a two-wire communication protocol widely used with sensors, EEPROMs, RTCs, and displays. The custom driver implements master-mode communication. I2C is a two-wire communication protocol widely used with sensors, EEPROMs, RTCs, and displays. The custom driver implements master-mode communication.

Driver Architecture

Applications access I2C through driver APIs. The driver generates START and STOP conditions, manages addressing, and handles data transfer. Applications access I2C through driver APIs. The driver generates START and STOP conditions, manages addressing, and handles data transfer.

Data Structures

i2c_config_t stores clock speed, address, and ACK settings. i2c_handle_t stores peripheral information and configuration. i2c_config_t stores clock speed, address, and ACK settings. i2c_handle_t stores peripheral information and configuration.

Component	Purpose
Configuration Structure	Stores settings
Handle Structure	Stores peripheral and config
APIs	Interface to application

Register Analysis

CR1 controls peripheral operation, CR2 configures clock information, OAR1 stores device address, DR handles data transfer, SR1/SR2 provide status information, CCR configures timing, and TRISE configures rise time. CR1 controls peripheral operation, CR2 configures clock information, OAR1 stores device address, DR handles data transfer, SR1/SR2 provide status information, CCR configures timing, and TRISE configures rise time.

API Analysis

i2c_init() configures the peripheral. i2c_generate_start() and i2c_generate_stop() control bus conditions. Data transfer is handled through master transmit and receive APIs. i2c_init() configures the peripheral. i2c_generate_start() and i2c_generate_stop() control bus conditions. Data transfer is handled through master transmit and receive APIs.

Application Example

The driver was tested using an LM75 temperature sensor connected through I2C1. Temperature data is read by STM32 and can be transmitted over USART. The driver was tested using an LM75 temperature sensor connected through I2C1. Temperature data is read by STM32 and can be transmitted over USART.

Advantages and Conclusion

The driver provides a reusable implementation of I2C communication and demonstrates understanding of sensor interfacing and serial protocols. The driver provides a reusable

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